Refined TAN et al. PBPK Model Evaluation & Extraction Questionnaire

Scope & Background: The scope of this template is to evaluate and extract PBPK models from scientific research papers using a refined version of the **Tan et al. PBPK Template** (which harmonizes OECD, US FDA, and EMA regulatory guidelines).

Key Refinements Applied for Research Papers

- **Elimination of Introduction Noise (Refined Section 2.2 & Routing Rules):** The original Tan et al. Section 2.2 ("Background information") was designed for regulatory dossiers and asks for general chemical exposure and toxicity summaries. When applied to a research paper, an LLM naturally scrapes the **Introduction**, pulling irrelevant historical manufacturing, epidemiological associations, and general toxicity studies not used in the model. In this refined template, general exposure/toxicity literature reviews are removed, duplicate items between Section 2.2 and Section 2.4.2 are consolidated, and the **Introduction** is strictly blocked (except its final paragraph stating the study objective).
- **Expanded Biological-Based ADME Determination & Model Assumptions (Refined Section 2.4.3):** The original Tan et al. Section 2.4.3 compressed structure, ADME mechanisms, and assumptions into a single paragraph. This template splits Section 2.4.3 into three explicit sub-sections (**2.4.3A Compartmental Structure**, **2.4.3B Biological-Based ADME Determination by Phase**, and **2.4.3C Model Assumptions & Justifications**) with structured prompts so the LLM extracts complete mechanistic explanations and mathematical implementations rather than vague fragments.
- **Deterministic Verification Instead of LLM Self-Scoring:** Replaces LLM self-reported confidence scores with 4 objective, post-extraction verification checks.

1. Model Identification & Reviewer Context

Field / Question	Allowed Paper Sections	Response / Inputs
1.1 Name of the PBPK Model & Target Compound(s) (Include parent chemical(s), active metabolites, and CAS/identifiers if stated)	Title, Abstract, Methods	[Enter response here]
1.2 Authors, Publication Year, Journal & DOI/PMID	Title Page / Header	[Enter response here]

Field / Question	Allowed Paper Sections	Response / Inputs
1.3 Date of Assessment & Reviewer / Pipeline Version	Reviewer Metadata	[Enter response here]

2. Refined TAN et al. PBPK Questionnaire (Sections 2.1 – 2.5.2)

Section 2.1 & 2.3: Executive Summary & Model Purpose

- **Strict Section Routing:** Extract ONLY from the **Abstract**, the **final paragraph of the Introduction** (study objective statement), and the **Discussion/Conclusion summary**. Do NOT extract general background paragraphs from the Introduction.

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
TAN 2.1.1	**2.1 Executive Summary**	**Rationale & Intended Purpose:** What is the specific scientific or regulatory rationale for conducting this PBPK analysis?	[Enter response here]
TAN 2.1.2	**2.1 Executive Summary**	**Model Overview & Scope:** Provide a high-level overview of the PBPK model, including modeled chemical(s), species, life stages/populations, exposure routes, and primary applications.	[Enter response here]
TAN 2.1.3	**2.1 Executive Summary**	**Summary of Key Conclusions:** What are the primary quantitative and scientific conclusions supported by the PBPK simulations?	[Enter response here]
TAN 2.3.1	**2.3 Model Purpose**	**Problem Formulation & Fit-for-Purpose Statement:** What exact scientific, dosimetry, or regulatory question(s) does the PBPK model address (e.g., cross-species extrapolation, gestational/lactational transfer, reverse dosimetry from biomonitoring, DDI prediction), and what internal dose metrics (e.g., Cmax, AUC,	[Enter response here]

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
		plasma/cord/milk/target-tissue concentrations) are predicted?	

Section 2.2 & 2.4.1–2.4.2: Modeling Strategy, Chemical Properties & Data Summary

- **Strict Section Routing:** Extract ONLY from **Materials and Methods** and **Tables/Figures**. Strictly EXCLUDE the **Introduction** and general toxicity/epidemiology background not used to parameterize, calibrate, or evaluate the model.

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
TAN 2.2.1	**2.2 & 2.4.2 Chemical Properties & Active Moiety**	**Model-Relevant Physicochemical & PK/TK Properties:** Which physicochemical and toxicokinetic/pharmacokinetic properties (e.g., molecular weight, logP, pKa, solubility, ionization, plasma protein binding fraction fu, blood-to-plasma ratio, half-life) are explicitly used in the Methods to build or parameterize the PBPK model? Specify the modeled chemical moiety (parent compound alone vs. parent plus coupled metabolites) and relevant exposure routes. *(Ignore general toxicity history in the Introduction).*	[Enter response here]
TAN 2.4.1	**2.4.1 Modeling Strategy**	**Step-by-Step Modeling Workflow:** What is the end-to-end modeling procedure from data preparation to model development, parameter calibration/fitting, model verification/evaluation, and model application? Explain how this workflow supports the stated model purpose.	[Enter response here]

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
TAN 2.4.2A	**2.4.2 Data Summary (In Vitro, In Silico & Disposition Studies)**	**Experimental & In Silico Studies Used for Parameterization:** Summarize the specific studies (with citations and experimental conditions such as species, cell/subcellular fraction, substrate concentrations, or QSAR software) used in the Methods to inform model structure and derive parameters: (1) mass balance or tissue distribution studies, (2) metabolism pathways and kinetic rates (Vmax, Km, CLint), and (3) active transporter kinetics (Tm, Kt) or protein binding measurements.	[Enter response here]
TAN 2.4.2B	**2.2 & 2.4.2 Calibration (Training) Datasets**	**Data Used for Model Calibration / Parameter Fitting:** List all in vivo animal, clinical, or human biomonitoring datasets (study citation, species/population, matrix, sample size, dosing/exposure context, and time points) used to calibrate, optimize, or fit unknown model parameters.	[Enter response here]
TAN 2.4.2C	**2.2 & 2.4.2 Evaluation (Validation) Datasets**	**Independent Data Used for Model Evaluation:** List all independent in vivo, clinical trial, or human biomonitoring datasets (study citation, species/population, matrix, sample size, and time points) held out from calibration and used to evaluate/validate the predictive capability of the model.	[Enter response here]

Section 2.4.3: Model Development, Compartmental Structure, Biological-Based ADME Determination & Assumptions

- **Strict Section Routing:** Extract ONLY from **Materials and Methods** (Model Structure, Physiology, and Parameter subsections), **Model Schematic Figures**, **Discussion** (limitations/assumptions), and **Supplementary Code**. Strictly EXCLUDE the **Introduction**.

2.4.3A. Model Lineage, Platform & Compartmental Structure

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
TAN 2.4.3.1	**2.4.3 Structure & Lineage**	**Model Lineage & Customized Modifications:** Does the PBPK model replicate a published model, refine/modify an existing model (provide exact citations of predecessor models), or introduce a brand-new model structure? Describe and justify all customized structural or physiological modifications made to any previous model.	[Enter response here]
TAN 2.4.3.2	**2.4.3 Structure & Platform**	**Compartmental Topology & Inter-Organ Connectivity:** If a generic PBPK platform is used, identify it. For each modeled species or life-stage sub-model (e.g., adult, gestation, lactation, fetus, infant), list all anatomical compartments, state the total compartment count, and describe how compartments are connected via blood/plasma flows (Qi), filtrate flows (Qfil), lymph flows, or inter-individual transfer pathways (placental, amniotic, lactational).	[Enter response here]

2.4.3B. Biological-Based ADME Determination (Mechanistic Basis by Phase)

- **Instruction:** Do not output short fragments. For each ADME phase below, explicitly extract: **(1) Biological/Physiological Mechanism**, **(2) Mathematical Implementation in the Model** (e.g., flow-limited vs. permeability/diffusion-limited, saturable Michaelis-Menten vs. first-order), **(3) Governing Parameters & Compartments**, and **(4) Exact Methods Page/Section**.

Item ID	ADME Phase	Refined LLM Extraction Prompt	Biological Mechanism	Mathematical Implementation in Model	Governing Parameters & Compartments	Source Location
TAN 2.4.3.3	**Absorption**	How does the model represent chemical uptake across all exposure routes (oral/GI, drinking water, breast milk suckling, inhalation, dermal, IV) and any presystemic/gut extraction?	[Enter mechanism]	[Enter mathematical model]	[Enter parameters & compar...	[Enter page/section]
TAN 2.4.3.4	**Distribution (Protein Binding & Blood Partitioning)**	How does the model represent plasma protein binding (e.g., albumin binding, constant free fraction fu vs. saturable binding), hematocrit (Htc), and blood-to-plasma partitioning?	[Enter mechanism]	[Enter mathematical model]	[Enter parameters & compar...	[Enter page/section]
TAN 2.4.3.5	**Distribution (Tissue Partitioning & Perfusion vs. Diffusion)**	Which tissues are modeled as perfusion-limited (flow-limited) vs. permeability/diffusion-limited, how are tissue:plasma partition coefficients (Pi or Kp) determined, and how are dynamic physiological volume/flow changes (e.g., pregnancy/growth) represented?	[Enter mechanism]	[Enter mathematical model]	[Enter parameters & compar...	[Enter page/section]
TAN 2.4.3.6	**Metabolism**	Is the compound metabolized or metabolically inert? If metabolized, which organs (liver, gut, etc.), enzymes, and rate laws (linear clearance vs. saturable Vmax/Km) govern parent biotransformation and metabolite formation?	[Enter mechanism]	[Enter mathematical model]	[Enter parameters & compar...	[Enter page/section]
TAN 2.4.3.7	**Excretion & Recirculation Mechanisms**	How is elimination represented (urinary, biliary/fecal, exhaled air)? What specific biological mechanisms explain long half-life or persistence (e.g., glomerular filtration of free chemical followed by saturable renal tubular reabsorption via OAT4/URAT1 using Tm and Kt, or enterohepatic	[Enter mechanism]	[Enter mathematical model]	[Enter parameters & compar...	[Enter page/section]

Item ID	ADME Phase	Refined LLM Extraction Prompt	Biological Mechanism	Mathematical Implementation in Model	Governing Parameters & Compartments	Source Location
		circulation), and how do they differ across life stages?				
TAN 2.4.3.8	**Barrier & Life-Stage Transfer (If Applicable)**	For gestational, lactational, or barrier models, what biological mechanisms and mathematical equations govern placental transfer (mother ↔ fetus), amniotic fluid exchange, and breast milk partitioning/suckling transfer (mother → milk → infant)?	[Enter mechanism]	[Enter mathematical model]	[Enter parameters & compar...	[Enter page/section]

2.4.3C. Explicit Model Assumptions, Justifications & Structural Impact

- **Instruction:** Extract every explicit simplifying assumption stated in the **Materials and Methods** or **Discussion**. For each assumption, provide the **Stated Assumption**, **Authors' Biological or Data-Gap Justification**, **Exact Mathematical/Structural Impact on the Model**, and **Source Location**.

Item ID	Assumption Category	Refined LLM Extraction Prompt	Stated Assumption	Authors' Biological or Data-Limitation Justification	Impact on Model Structure / Equations	Source Location
TAN 2.4.3.9	**1. Tissue Perfusion / Diffusion & Lumping**	What assumptions are made regarding well-stirred (flow-limited) behavior, organ lumping (e.g., Rest of Body or 1-compartment infant), or omitted compartments (e.g., omitting mammary tissue in lactation)?	[Enter assumption]	[Enter justification]	[Enter impact on equations/s...	[Enter page/section]
TAN 2.4.3.10	**2. Protein Binding & Free Fraction Constancy**	What assumptions are made regarding plasma/tissue protein binding, red blood cell partitioning, and the constancy of the unbound fraction (fu or Free) across concentrations, species, or life stages?	[Enter assumption]	[Enter justification]	[Enter impact on equations/s...	[Enter page/section]
TAN 2.4.3.11	**3. Cross-Species & Cross-Chemical Parameter Borrowing**	Which parameters (e.g., tissue partition coefficients Pi, amniotic transfer rates, transporter constants) are assumed to be equal to values measured in	[Enter assumption]	[Enter justification]	[Enter impact on equations/s...	[Enter page/section]

Item ID	Assumption Category	Refined LLM Extraction Prompt	Stated Assumption	Authors' Biological or Data-Limitation Justification	Impact on Model Structure / Equations	Source Location
		surrogate species (rat/mouse/monkey) or analogue chemicals, and why?				
TAN 2.4.3.12	**4. Clearance, Resorption & Ontogeny**	What assumptions are made regarding zero metabolism, active renal reabsorption (Tm, Kt), or immature/omitted transporter activity in fetuses and infants?	[Enter assumption]	[Enter justification]	[Enter impact on equations/s...	[Enter page/section]
TAN 2.4.3.13	**5. Dosing, Exposure & Steady-State Initialization**	What assumptions are made regarding pre-pregnancy steady-state run-in duration, route surrogates (e.g., continuous IV input representing dietary/oral intake), caloric/fluid intake adjustments during lactation, or milk production equaling suckling rate?	[Enter assumption]	[Enter justification]	[Enter impact on equations/s...	[Enter page/section]

Section 2.4.4 – 2.4.7: Equations, Parameters, Simulations & Software

- **Strict Section Routing:** Extract from **Materials and Methods**, **Input Parameter Tables (e.g., Tables 1–3)**, **Table Footnotes**, and **Supporting Information**. Exclude sensitivity output tables and reverse-dosimetry result tables from input parameter extraction.

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
TAN 2.4.4	**2.4.4 Model Equations**	**Mathematical, Dynamic Growth & Allometric Scaling Equations:** Provide all mathematical equations defining the PBPK model so it can be replicated in another software package: (1) differential mass-balance equations (dAi/dt), (2) saturable transport or metabolic rate equations, (3) time-varying physiological growth equations (e.g., maternal plasma, fat, placenta, fetal weight), and (4)	[Enter response here]

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
		allometric scaling formulas (e.g., BW^0.75 or BW^-0.25).	
TAN 2.4.5A	**2.4.5 Model Parameters (Tabulation & Scaling)**	**Complete Input Parameter Extraction:** Report all physiological and chemical-specific parameters from both tables (including table footnotes) and Methods text (such as water/food intake rates). Include parameter name, symbol, value (or dynamic_start_value and dynamic_end_value for time-varying gestational/lactational ranges), units, allometric scaling exponent in units (e.g., L/h/kg^0.75), compound, species, and life stage.	[Enter response here or link ...
TAN 2.4.5B	**2.4.5 Parameter Sources & Competing Values**	**Parameter Provenance, Original Citations & Competing Values:** For each parameter, specify its source category (Measured in vitro/in vivo, Fitted/Calibrated to observed data, Literature, In silico QSAR, Software database, or Assumed) and cite the **original primary publication** (author, year, page/table) rather than a secondary citation. If a parameter has competing values in the literature (e.g., alternative human half-lives or alternative placental transfer rates), report all evaluated values, the chosen value, and the rationale.	[Enter response here]
TAN 2.4.5C	**2.4.5 Parameter Distributions & Uncertainty**	**Probability Distributions & Uncertainty Bounds:** If probabilistic simulations are performed, list parameter probability distribution types	[Enter response here]

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
		(normal, log-normal, triangular, uniform) and distribution parameters (mean, SD/variance, 95% CI, truncated range). If deterministic, state how physiological time-varying ranges or alternative parameter bounds are parameterized.	
TAN 2.4.6A	**2.4.6 Model Simulations (Exposure & Dosing)**	**Exposure & Administration Conditions:** Provide full details of all simulated dosing/exposure scenarios: routes of administration, timing and duration of exposure (e.g., 30-year pre-pregnancy run-in, gestation duration, lactation duration), formulation/vehicle, feeding/fasting state, administered doses or environmental concentrations (e.g., water ppb), and physiological intake rates (e.g., mL/kg/day water intake, ventilation rate, milk suckling rate) required to compute apparent doses.	[Enter response here]
TAN 2.4.6B	**2.4.6 Model Simulations (Samples, Subjects & Trials)**	**Simulated Matrices, Virtual Population & Probabilistic Setup:** Specify: (1) biological sample types (e.g., parent chemical or metabolite in maternal plasma, cord blood, breast milk, amniotic fluid, urine) and collection times, (2) characteristics of test subjects and simulated virtual populations (species, N, sex, ethnicity, age at conception/delivery, BW trajectories, healthy vs. disease state), and (3) number of trials, virtual subjects per trial, or Monte Carlo/MCMC iterations.	[Enter response here]

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
TAN 2.4.7	**2.4.7 Software & Numerical Solvers**	**Software Platform, Version, Solvers & Code Availability:** State the software name and version, product website/manual reference, ordinary differential equation (ODE) solver algorithm, interpolation method (e.g., linear TABLE interpolation), optimization/statistical algorithms, and whether executable model code is provided in the Supporting Information.	[Enter response here]

Section 2.5.1 – 2.5.2: Model Evaluation & Sensitivity / Uncertainty Analyses

- **Strict Section Routing:** Extract from **Materials and Methods** (Evaluation and Sensitivity subsections), **Results**, **Discussion**, and **Evaluation/Sensitivity Figures & Tables (e.g., Figures 3–7, Tables 4–5)**.

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
TAN 2.5.1A	**2.5.1 Model Evaluation (Training vs. Validation Fit)**	**Comparison of Predictions with Observed Data (Training vs. Evaluation):** Summarize how model predictions compare against observed in vivo time-concentration data across figures and tables. Explicitly distinguish results showing model fit to the **training (calibration) dataset** from results demonstrating predictive performance on **independent evaluation datasets**.	[Enter response here]
TAN 2.5.1B	**2.5.1 Goodness-of-Fit & Applicability**	**Goodness-of-Fit Metrics & Model Limitations:** Describe the statistical tests, fold-error criteria (e.g., predicted-to-observed ratio within a factor of 2), or visual inspection criteria used to assess	[Enter response here]

Item ID	TAN Section	Question & Refined LLM Extraction Prompt	Response / Extracted Details
		goodness-of-fit, interpret the results, and discuss model applicability and limitations for the intended purpose.	
TAN 2.5.2A	**2.5.2 Sensitivity Analysis Protocol**	**Sensitivity Analysis Type, Sampling & Calculation Method:** Specify: (1) type of sensitivity analysis (local vs. global), (2) sampling approach (one-at-a-time perturbation percentage such as 1% vs. all-at-a-time; steady state vs. specific gestational/lactational time points; dose levels tested), (3) mathematical method (normalized sensitivity coefficient formula, Morris, eFAST), and (4) reporting threshold/cutoff criterion (e.g., absolute SC > 0.07 or >= 1.0).	[Enter response here]
TAN 2.5.2B	**2.5.2 Key Sensitive Parameters & Uncertainty Impact**	**Influential Parameters & Overall Reliability:** Provide an overview of the key sensitive parameters identified across matrices/time points and state the overall impact of parameter sensitivity, variability, and uncertainty (e.g., alternative half-life values or intake ranges) on model outputs and reliability.	[Enter response here]

3. Explanation of Confidence Score & Deterministic Rule-Based Verification

Why Current Confidence Scores Are Unreliable

In the current extraction pipeline, the confidence scores are **LLM self-reported values and hardcoded schema defaults** rather than independent quality checks:

- LLM Self-Scoring in Prompt (STEP 6 — CONFIDENCE) and Structure Tabs: The LLM generates its own confidence float (0.98 for all compartments and 0.90–0.95 for all

connections) or label (HIGH, MEDIUM, LOW) inside the same JSON call that extracts the data. When the LLM makes a structural error (such as wiring maternal_plasma directly to placenta via ktrans1C/ktrans2C and leaving fetal_plasma disconnected), it still scores its own extraction 0.90.

- Static Default Labels in Parameter Tabs: Across all 102 rows of Model Parameters, the six confidence columns (value_confidence, unit_confidence, species_confidence, formulation_confidence, parameter_class_confidence, provenance_confidence) display the identical static pattern (high, high, high, medium, high, high, needs_review = False), even when value is NaN (on all 25 time-varying range rows), unit_standardized is NaN, or reference is NaN.

Deterministic Post-Extraction Verification Score (Replacing LLM Self-Scoring)

Remove confidence from the LLM prompt and calculate a Deterministic Verification Score (0.00 to 1.00) in Python post-processing using 4 automated checks:

Check ID	Automated Python Verification Rule	Weight	Failure Action
Check 1	**Verbatim Source Match:** Both `symbol_original` and `value_text` match verbatim via string/regex search inside the extracted text/table of `source_page`.	**+0.25**	Flag `needs_review = True` (Potential hallucination or symbol mismatch)
Check 2	**Numeric Parse Completeness:** Either a point `value` OR both `dynamic_start_value` and `dynamic_end_value` (for time-varying ranges) are populated as valid floats (not `NaN`/null).	**+0.25**	Flag `needs_review = True` (Unparsed range or footnote string)
Check 3	**Unit & Dimensional Consistency:** `unit_standardized` is non-null and matches the expected ontology dimension for `parameter_subclass` (e.g., `unitless` for partition coefficients and `Free`; `L/h/kg^0.75` for clearance/Tm).	**+0.25**	Flag `needs_review = True` (Missing or mismatched unit)

Check ID	Automated Python Verification Rule	Weight	Failure Action
Check 4	**Provenance & Original Citation Completeness:** Both the paper location (`Table 1, Methods, p. 11`) AND the original literature citation (`reference`, e.g., Brown et al. 1997) plus `parameter_status` are populated.	**+0.25**	Flag `needs_review = True` (Missing original reference citation)